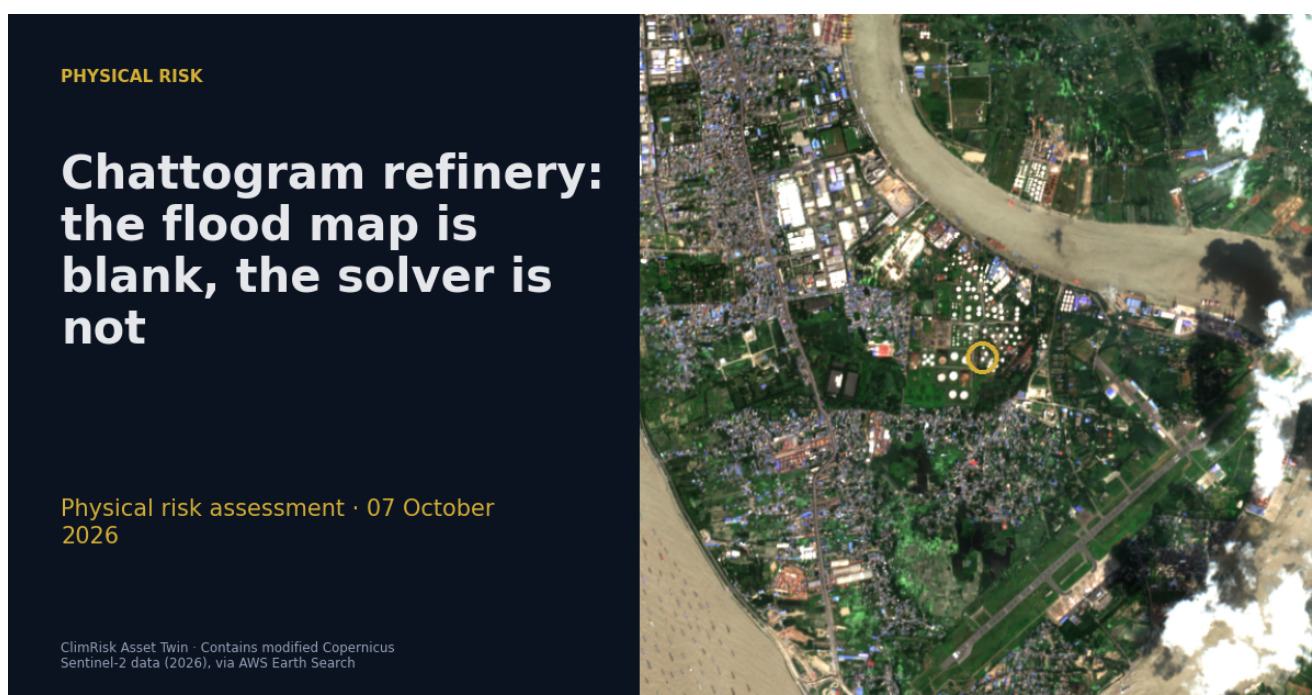


Chattogram refinery: the flood map is blank, the solver is not

64 structures modelled on real footprints and terrain. In the worst mapped flood (River flood, 1-in-10) a 2D solver puts water on 4 of them, 4 with equipment under water, up to 4.7 m deep, on a site where the hazard map shows 0.0 m. 39 exceed their assumed design wind in the cyclone wind, 1-in-250 years scenario.

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1991. The strongest storm in the track archive within reach of Chattogram refinery was an unnamed tropical system: modelled sustained wind of 67 m/s at the site (Cat 4), with the centre 5 km away at its closest [1]. The archive holds 20 such passages since 1980. That is the loud part of the record. It is not, on its own, a statement about what the site can take.

Chattogram refinery in six numbers

What the engine found, measured at the site

64

structures modelled on mapped footprints

OpenStreetMap + 9 m terrain

4.7 m

deepest solver water on the site

River flood, 1-in-10

4 of 39

structures reached in the worst flood

2D shallow-water solver

67 m/s

strongest storm on record (1991)

IBTrACS best tracks, Holland model

1.63%

modelled annual damage, share of value

JRC / Emanuel curves × hazard

6

of 6 findings with a verified fix

solver / re-evaluation

Source: ClimRisk Asset Twin; IBTrACS; JRC CEMS-GloFAS; Deltares; OpenStreetMap; AWS Terrain Tiles

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Every number is computed by the engine for this site; the source is stamped on the image.

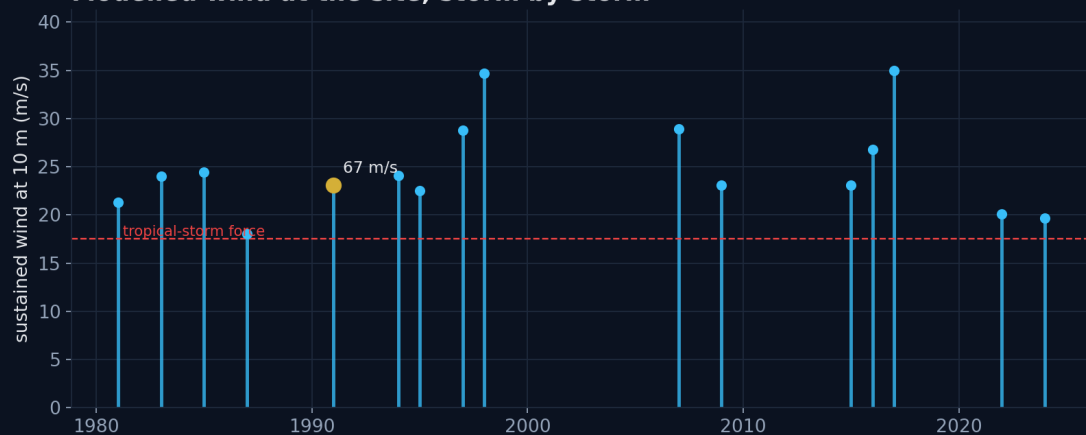
4 of 39 structures reached by water

The loud risk is what the maps show: a flood layer with 0.0 m on the site, and a cyclone record whose worst passage brought 67 m/s. The quiet one is what the site does with the water. The solver finds low ground the 90 m map cannot resolve and puts water on 4 of 39 structures, 4 of them with equipment under water. A refinery is a tank farm, a set of process units, a jetty and a flare, tied together by pipe racks that nobody wants to disturb.

What has already happened here

The track archive holds 20 tropical storm or stronger passages within reach of this site since 1980 [1]. The strongest, Unnamed storm in 1991, brought about 67 m/s (Cat 4) at the site, passing 5 km away.

Modelled wind at the site, storm by storm



Source: NOAA IBTrACS v04r01; Holland (1980) wind field; ClimRisk engine

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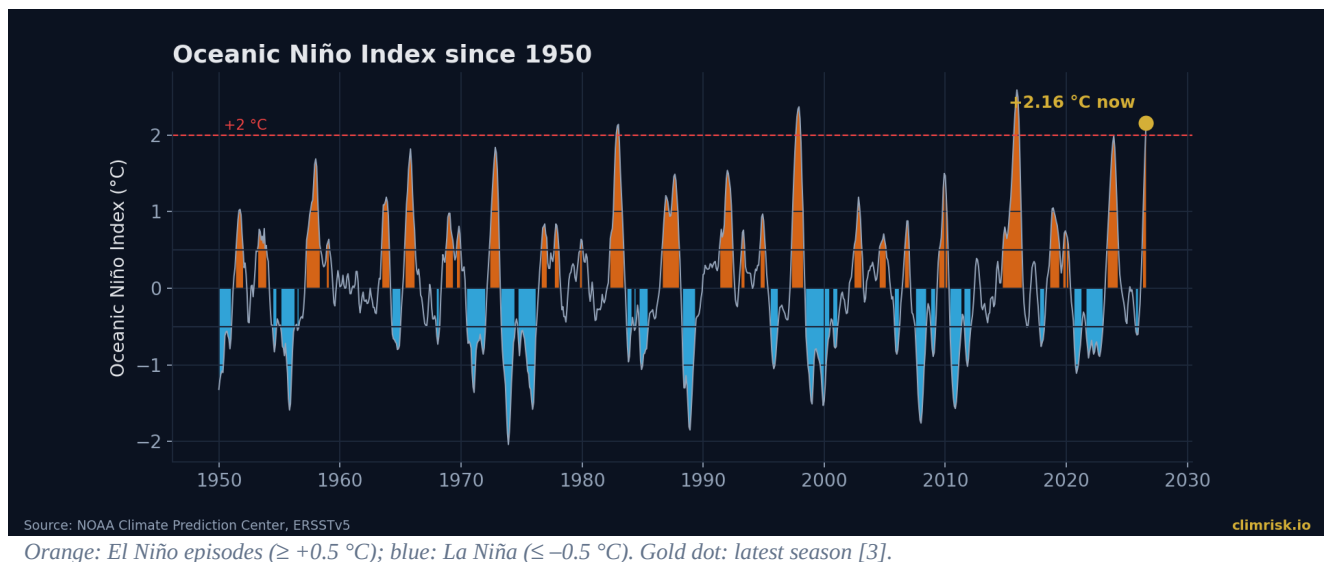
Each stem is one storm's peak modelled wind at the site. Gold: the strongest [1].

Storm	Year	Peak wind at site	Category	Closest approach
Unnamed storm	1991	67 m/s	Cat 4	5 km
Unnamed storm	1997	59 m/s	Cat 4	14 km
Mora	2017	35 m/s	Cat 1	20 km
Unnamed storm	1998	35 m/s	Cat 1	33 km
Sidr	2007	30 m/s	tropical storm	145 km
Akash	2007	29 m/s	tropical storm	44 km

The all-cause record (reanalysis, 1950 onward) adds non-cyclone wind. The five strongest daily gusts: 1991-04-29 42 m/s; 1997-05-19 33 m/s; 2017-06-12 32 m/s; 1966-10-01 32 m/s; 1995-11-25 31 m/s [2]. Reanalysis smooths peaks, so these are floors, not ceilings.

No operator-reported incidents were supplied. Adding them lets the replay be checked against what actually happened, which is the best test of any of the numbers here.

The background state matters for the season. The Oceanic Niño Index for JAS 2026 stands at +2.16 °C (El Niño); since 1950 it has crossed +2 °C on only 4 occasions [3]. ENSO shifts cyclone and flood odds in many basins, but the engine does not condition its return periods on it: treat the index as context, not as an adjustment to the figures below.



The loud risk, priced

Cyclone wind is the risk everyone plans for. Here the archive implies 0.44 tropical-storm-force passages a year and a 1-in-100-year wind of 61 m/s (Cat 4) [1]. Run through the damage curve [4] against an asset value of \$500.0m (user-supplied value), the modelled average annual damage is **1.63% of value** (\$8.1m a year), and a 1-in-100-year event year costs **\$219.1m**, 44% of value.



The flood maps are the other loud number. River and coastal layers are undefended; the engine credits a protection standard of about 1-in-50 (river) and 1-in-50 (coast) from FLOPROS [5]. Mean mapped depth over the site reaches 0.00 m in the deepest scenario [6][7].

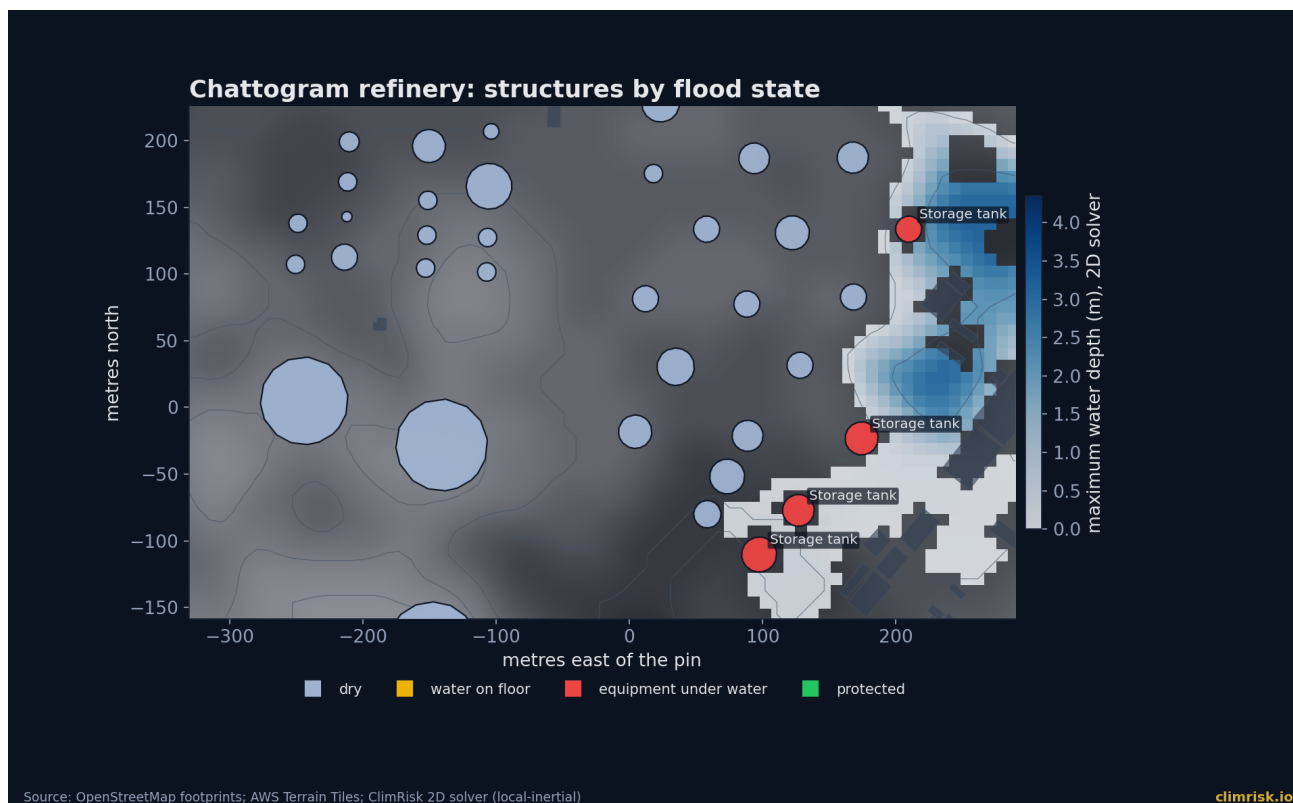
All 17 mapped flood scenarios (river 1-in-10 to 1-in-500, coastal surge today and with sea-level rise) put 0.00 m of water on the site. That is the loud number, and it is zero.

Wind scenario	Sustained at 10 m	Category
Cyclone wind, 1-in-10 years	34 m/s	Cat 1
Cyclone wind, 1-in-50 years	53 m/s	Cat 3
Cyclone wind, 1-in-100 years	61 m/s	Cat 4
Cyclone wind, 1-in-250 years	72 m/s	Cat 5
All-cause wind (ERA5), 1-in-10 years	21 m/s	
All-cause wind (ERA5), 1-in-50 years	26 m/s	
All-cause wind (ERA5), 1-in-100 years	28 m/s	

Return-period winds from the cyclone model and the all-cause reanalysis fit [1][2].

The quiet risk: what the site does to the water

A refinery is a tank farm, a set of process units, a jetty and a flare, tied together by pipe racks that nobody wants to disturb. The maps stop at a pixel 90 m wide. The site does not. At Chattogram refinery the twin carries 64 structures on their mapped footprints, each with a floor level taken from 9 m terrain, and a 2D solver moves the water between them. The quiet risk is an empty floating-roof tank that lifts when the bund fills, shearing the pipework it was connected to.



Maximum water depth in the worst scenario (River flood, 1-in-10), with each structure coloured by what the water does to it. Blue shading: solver depth; contours: terrain [8][9][10].

Solver check: mass residual 0.0%, solver wet land 21% against 0% on the map (the solver floods more land than the 90 m map: it finds low pockets and channels the map cannot see, or the terrain tiles sit lower than the map's terrain. Treat these depths as a low-lying-ground warning, not as map-validated depths.).

The flood maps show no depth on the site itself, yet the 2D solver reaches structures. The solver takes its water level from the nearest wet map pixels (up to 600 m away) and routes it over 9 m terrain, which can fill low pockets that the 90 m flood map smooths over. Read the flood findings below as a warning to check ground levels on site, not as a mapped flood.

River flood, 1-in-10: 4 component(s) reached by water

Structure	Kind	Floor above datum	Water depth	State
Storage tank 28	tank	6.68 m	2.26 m	equipment inundated
Storage tank 14	tank	8.20 m	0.35 m	equipment inundated
Storage tank 12	tank	6.44 m	0.06 m	equipment inundated
Storage tank 10	tank	6.17 m	0.04 m	equipment inundated

Worst-reached structures. Floor levels come from the terrain model plus an assumed plinth until the operator supplies as-built values.

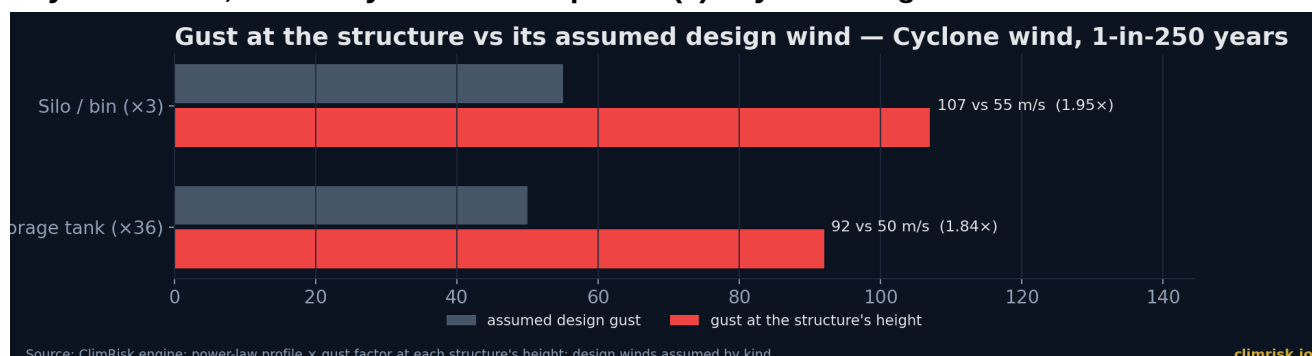
Why: The river flood map puts the water surface at about 6.84 m above datum at the edge of the site model (range 6.84–8.33 m). The solver routes that water over the terrain around the buildings; it reaches the first affected structure after about 95 min of the rise. The affected structures stand at 6.0–8.1 m above datum with floors 0.15 m above ground unless edited, so water gets in before the equipment inside is lifted clear.

What it costs: Floating tanks, damaged pipework and flooded utilities cause releases and unit trips; flooded control and power systems force a controlled shutdown, and terminals and roads are cut.

River flood, 1-in-100: 4 component(s) reached by water. Storage tank: water 0.04–2.26 m over the floor, equipment under water (×4)

Coastal surge + tide, 1-in-100, with sea-level rise to 2050: 2 component(s) reached by water. Storage tank: water 0.09 m over the floor, equipment under water (×2)

Cyclone wind, 1-in-250 years: 39 component(s) beyond design wind



A 72 m/s sustained wind at 10 m becomes a larger gust at the top of a tall structure, and load grows with the square of speed. Red bars exceed the assumed design gust.

Why: A 71.9 m/s sustained wind at 10 m becomes a gust of up to 107.1 m/s at the top of the tallest affected structure (wind rises with height), and load grows with the square of speed.

What it costs: Floating-roof tanks, flare stacks, columns and pipe racks are wind-sensitive. Storm surge combined with wind threatens jetties and marine loading, and power loss forces an emergency shutdown.

Design winds here are assumed by structure kind (tank, silo, building). Replace them with the as-built design values in the twin editor: that is the single most valuable edit for this section.

Finding	Hazard	Structures	Verified fix
River flood, 1-in-10: 4 component(s) reached by water	flood	4	yes
River flood, 1-in-100: 4 component(s) reached by water	flood	4	yes
Coastal surge + tide, 1-in-100, with sea-level rise to 2050: 2 component(s) reached by water	flood	2	yes
Cyclone wind, 1-in-100 years: 39 component(s) beyond design wind	wind	39	yes
Cyclone wind, 1-in-250 years: 39 component(s) beyond design wind	wind	39	yes
Unnamed storm 1991 (strongest on record at the site): 39 component(s) beyond design wind	wind	39	yes

Every finding the Asset Twin agent raised, with whether a measure re-run through the solver or structure check resolves it.

River flood, 1-in-10, River flood, 1-in-100: the solver floods more land than the 90 m map: it finds low pockets and channels the map cannot see, or the terrain tiles sit lower than the map's terrain. Treat these depths as a low-lying-ground warning, not as map-validated depths.

What protects it, ranked and verified

Structures reached by water, before and after (River flood, 1-in-10)



Source: ClimRisk 2D solver / component re-evaluation; measures proposed and verified by the engine

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Each bar re-runs the structure check with the measure applied. The barrier is re-run through the solver with the wall in the model.

Hazard	Measure	What remains	Verified by
flood	raise flood-critical equipment 2.6 m	0 equipment inundated, 4 wet floor	re-evaluation of component levels against the solver depths
flood	perimeter flood barrier 3.8 m	1 equipment inundated, 0 wet floor	2D solver with the wall in the model
wind	wind retrofit to 120 m/s	0 still beyond design	re-evaluation

Recommended: flood barrier 3.8 + wind retrofit 120.0. With it no equipment is inundated and nothing is beyond design wind in the base scenarios; it still holds up to +1.0 m of extra water level and +10% wind speed.

Solution	Resolves base scenarios	Flood margin	Wind margin
flood barrier 3.8 + wind retrofit 120.0	yes	+1.0 m	+10%
raise equipment 2.6 + flood barrier 3.8 + wind retrofit 120.0	yes	+1.0 m	+10%
raise equipment 2.6 + wind retrofit 120.0	no	+0.0 m	+10%
wind retrofit 120.0	no	+0.0 m	+10%
flood barrier 3.8	no	+1.0 m	+0%
raise equipment 2.6 + flood barrier 3.8	no	+1.0 m	+0%

Scenario lab: every mapped flood scenario shifted by +0 to +1.0 m and every wind scaled $\times 1.0$ to $\times 1.2$. Ranked by physical performance; no costs are invented. Flood margins shift the solver's depth field (screening-grade); open the 3D twin and drag the water-level slider to re-run the solver.

What this means, by reader

For the operator. Start with: river flood, 1-in-10: 4 component(s) reached by water. The cheapest verified fix is listed above; edit the twin with as-built values and the findings change with it.

For the lender or insurer. 6 scenario(s) produce a finding at this site, and the flood map alone would have shown none. Ask which verified measures are in the capital plan and when. The hazard maps are undefended; local defences are credited only at the FLOPROS standard unless the operator supplies better.

For the investor. Direct physical loss is 1.63% of value a year on average, and \$219.1m in a 1-in-100 year. These are direct-damage figures. A site that stops for weeks costs more than the equipment it loses, and that is not in these numbers.

Method and limitations

Flood depths come from a 2D shallow-water solver (local-inertial form [10]) on 9 m terrain [8] with every building a solid obstruction, forced with the water level implied by the flood map [6][7]. Event duration is assumed; rainfall, drains, pumps and waves are not modelled. Wind at a structure's height uses a power-law profile and a gust factor, and design winds are assumed by kind until edited. Tornado and downburst are not modelled. Footprints come from OpenStreetMap [9]; unmapped process equipment must be added with the twin editor.

- 64 component heights are assumed by kind (no OSM height tag).
- All equipment flood-critical levels are assumed by kind unless edited (building, silo, tank).
- 64 components use an assumed design gust by kind; enter the as-built design wind speed to replace it.
- Finished floor = ground + 0.15 m unless edited; gust factor 1.25; wind profile exponent 0.16.

How ClimRisk measured this

- Terrain and footprints → a 3D twin of 64 structures on 300 m radius.
- Hazards → JRC river, Deltares coastal, IBTrACS cyclone and ERA5 all-cause wind, each cited.
- Quantify → a 2D hydraulic solver per flood scenario and a height-adjusted gust check per structure.
- Protect → measures proposed by the engine and re-run through the solver or the structure check before they are reported.
- Price → damage curves and the analyst's Monte Carlo for the loss figures. Every number above can be regenerated from the twin.

Sources

1. NOAA IBTrACS tropical-cyclone best tracks — 1980–2025, Holland wind model <https://www.ncei.noaa.gov/products/international-best-track-archive>
2. ERA5 reanalysis via Open-Meteo archive — daily maximum gust, 1950 onward <https://open-meteo.com/en/docs/historical-weather-api>
3. NOAA Climate Prediction Center, Oceanic Niño Index (ERSSTv5) — monthly 3-month running mean, 1950 to date https://www.cpc.ncep.noaa.gov/products/analysis_monitoring/ensostuff/ONI_v5.php
4. Emanuel (2011), Global warming effects on U.S. hurricane damage, *Weather, Climate & Society* 3:261 — wind damage sigmoid <https://doi.org/10.1175/WCAS-D-11-00007.1>
5. Scussolini et al. (2016), FLOPROS: an evolving global database of flood protection standards, *NHESS* 16:1049 — protection standard, as a return period <https://doi.org/10.5194/nhess-16-1049-2016>
6. JRC CEMS-GloFAS river flood hazard maps v2.1 — undefended 3 arc-second depth maps <https://data.jrc.ec.europa.eu/collection/id-0054>
7. Deltares Global Flood Maps v2021.06 (GTSMip6 surge + tide) — undefended coastal depth maps <https://deltares-research.github.io/global-flood-maps/>
8. AWS Terrain Tiles (Terrarium), ~9 m <https://registry.opendata.aws/terrain-tiles/>
9. OpenStreetMap contributors — building and industrial footprints — ODbL <https://www.openstreetmap.org/copyright>
10. Bates, Horritt & Fewtrell (2010), local-inertial shallow-water formulation, *J. Hydrology* 387:33 <https://doi.org/10.1016/j.jhydrol.2010.03.027>

Produced by the ClimRisk engine from the sources listed. Model outputs with the limitations stated; not forecasts or investment advice.